

Colloquium for Bachelor thesis

How do AI-supported monitoring systems work as digital ones
Information systems and to what extent they enable data-based analysis
and influencing behavior?

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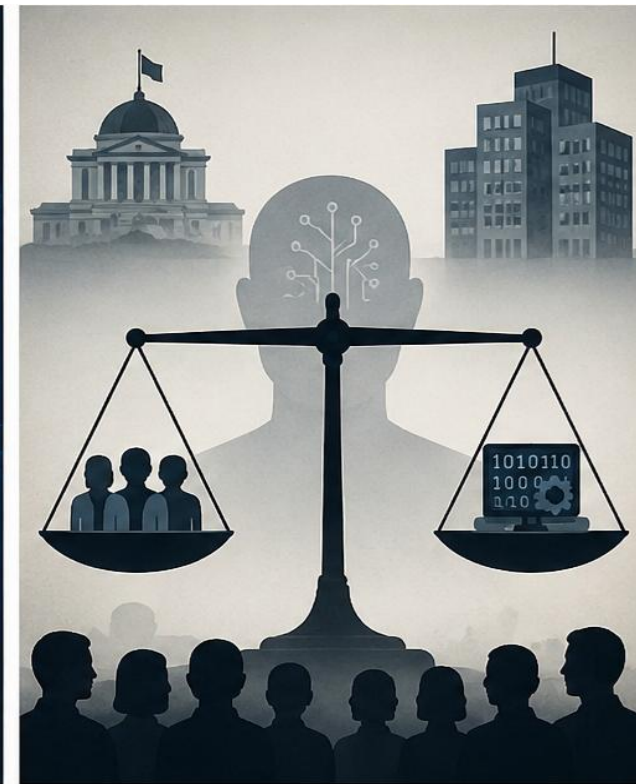
Increasing digitalization and data-driven platforms

- Data-driven platforms
- Continuous data collection



AI analyzes and evaluates human behavior

- Analysis of usage behavior
- Pattern recognition
- Forecasts and assessments



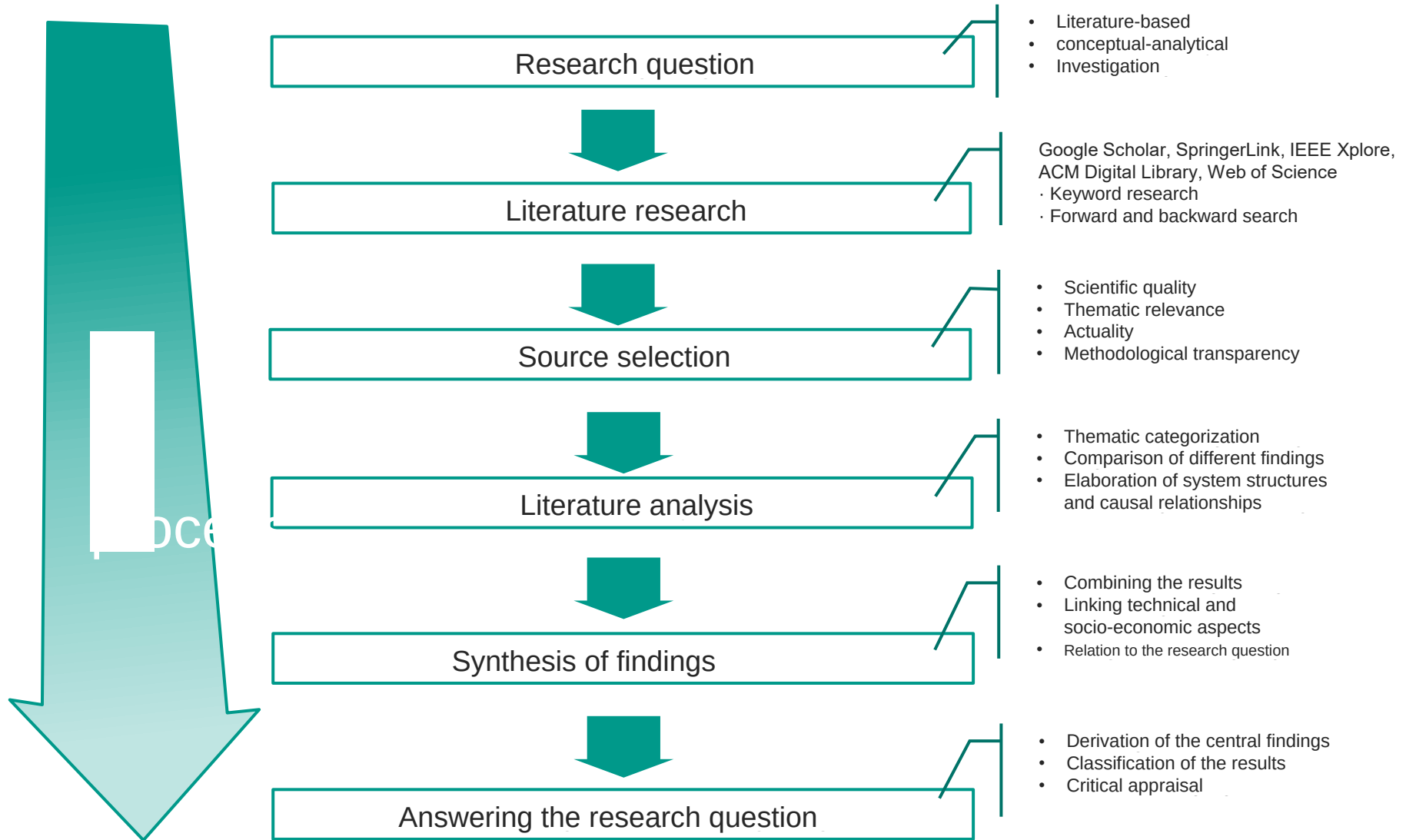
Social relevance

algorithmic decisions

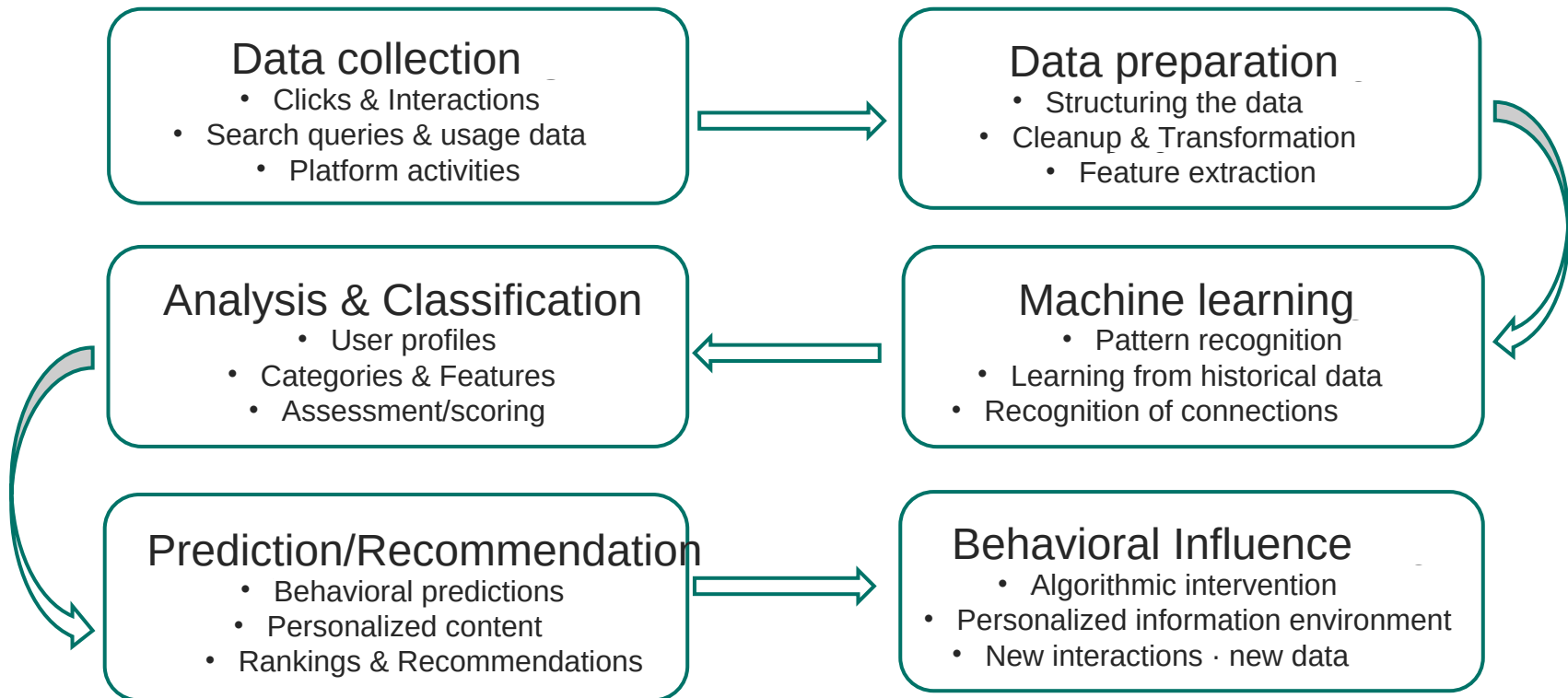
- Algorithmic decisions
- Personalization and control
- Transparency and control

Sources:

See K. C. Laudon & J. P. Laudon 2018, pp. 56-58; See Zuboff 2019, pp. 14-16; See Kitchin 2014, p. 722.

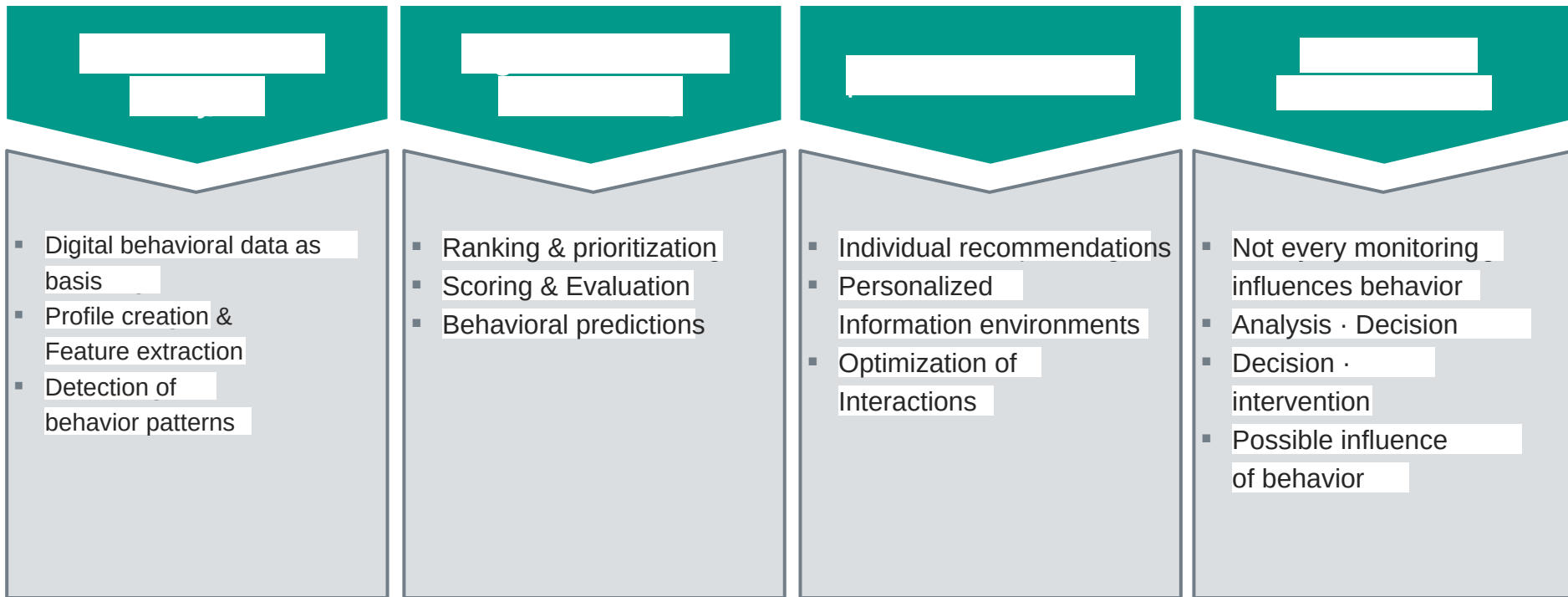


3. How an AI-supported monitoring system works



Sources:
See Goodfellow et al. 2016, pp. 4-5, 104-105; See K. C. Laudon & J. P. Laudon 2018, pp. 254-255, 259.

4. Key results



Central insight:
Behavioral data analysis can be done via algorithmic assessments and personalized decisions through to possible behavior influencing lead.

Sources:
See Kitchin 2014, p. 722; See Adomavicius & Tuzhilin 2005, pp. 734-735; See Zuboff 2019, pp. 14, 66-68.

5. Critical appraisal

Limits of technological analysis

- Proprietary systems
- Limited transparency

Limits of the literature base

- Heterogeneous evidence
- Limited generalizability

Methodological limitations

- No own primary data
- No empirical testing of the causal relationships



Conceptually explanatory · empirically limited

Sources:

See Pasquale 2015, pp. 6-9, 14-15; See Orben & Przybylski 2019, pp. 2, 4-5.

6. Conclusion & Outlook

Conclusion

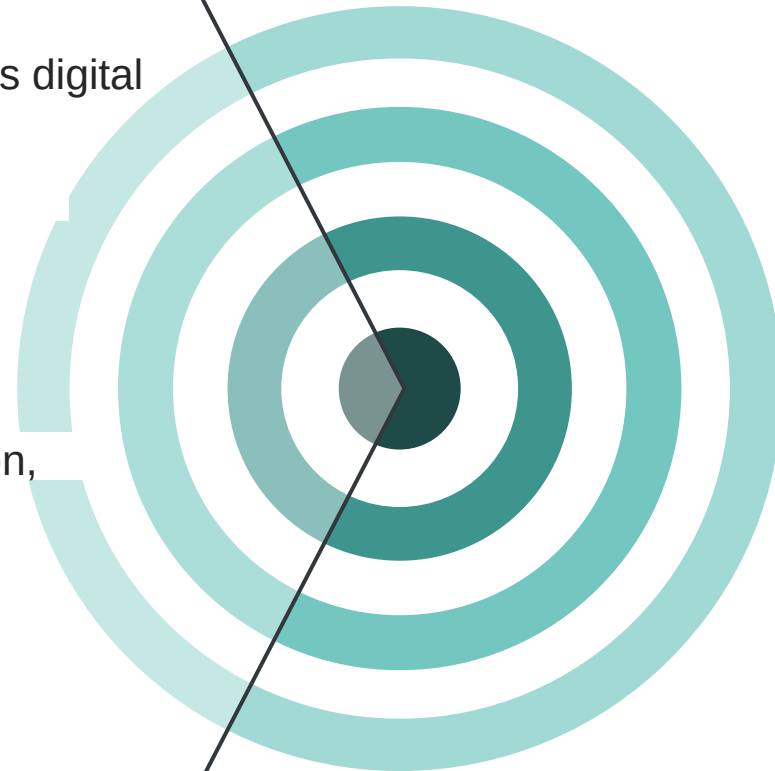
- Data · Analysis · Evaluation · Decision
- AI-supported monitoring systems can be described as digital Classify information systems
- Possible behavioral influence through algorithmic Interventions
- Sociotechnical and societal impacts

Key message:

AI-supported monitoring systems combine data collection, Analysis and algorithmic decision making and can thereby influence behavior based on data.

outlook

- More transparency and traceability
- Further development of governance and regulation
- Empirical investigation of concrete systems



A background image of two business professionals shaking hands in a modern office setting. The image is overlaid with a light blue grid pattern. Several circular icons are scattered across the image: a person in a hard hat at the top center, a person in a suit at the top right, a person in a hard hat at the bottom center, a person in a hard hat at the bottom left, and a person in a hard hat at the bottom right. There are also some abstract line drawings of people and objects.

Thank you for your attention!

Do you have anymore questions?